EXECUTE 5.O HACKATHON 2026

Theme: Innovation for Sustainable Development

PPT (Ideation) Submission Round Brochure

Organised by:
Entrepreneurship Cell
Delhi Technological University

Tracks:

- Sustainable Development
- Innovation & Emerging Technologies

Submission Deadline:
26 January 2026 | 11:59 PM

THEME & TRACK DETAILS

Innovation for Sustainable Development

Participants are expected to propose solutions that leverage **technology, data, and intelligence** to solve sustainability challenges related to resource management, efficiency, and governance.

Tracks:

Sustainable Development

Focuses on:

- Resource optimization
- Environmental impact reduction
- Long-term sustainability

Innovation & Emerging Technology

Focuses on:

- Advanced algorithms
- AI/ML-driven systems
- Scalable and adaptive digital solutions

★ Teams may approach the problem from **either or both tracks**.
★ Winner will be declared from the overall comparison of both tracks. No separate winner for any track.

PROBLEM STATEMENT

Problem Title

AI-Driven Adaptive Sustainability Operating System (ASOS) for Decentralized Communities

---

Problem Category

Software / Coding-Based Problem

Background & Context

Modern campuses, residential societies, and urban communities consume large quantities of critical resources such as electricity and water. Existing sustainability solutions primarily focus on passive monitoring, lacking predictive intelligence, decision support, and adaptability to human behavior.

Fragmented data, uncertain demand patterns, and lack of policy simulation tools prevent administrators from making informed, long-term sustainable decisions.

There is a critical need for an intelligent, adaptive, and scalable Sustainability Operating System that functions as a decision-making engine, not merely a reporting dashboard.

PROBLEM OBJECTIVE & SCOPE

Objective

Design a software-based Sustainability Operating System capable of:

- Integrating multi-source sustainability data
- Predicting future resource demand and stress points
- Recommending optimized actions
- Simulating sustainability policies before implementation
- Balancing environmental impact, cost, and user comfort

Scope of the Solution

The proposed solution should be applicable to:

- University campuses
- Residential societies
- Smart city wards
- Decentralized communities

The system must remain scalable, explainable, and feasible under real-world constraints.

FUNCTIONAL REQUIREMENTS

---

1. Data Ingestion & Management

- Handling of time-series data from heterogeneous sources
- Management of missing, delayed, or noisy data
- Data validation and preprocessing mechanisms

---

2. Predictive & Prescriptive Intelligence

- Demand forecasting and peak load prediction
- Detection of inefficiencies and anomalies
- Recommendation of optimized actions considering multiple objectives

---

3. Policy Simulation & Scenario Analysis

- Support for “what-if” simulations such as:
  - Pricing changes
  - Usage limits
  - Incentive mechanisms

- Quantified outcomes for environmental and economic impact.

HUMAN & INNOVATION REQUIREMENTS

---

4. Human-Centric Sustainability Design

- Modeling of user behavior and compliance
- Adaptive feedback mechanisms
- Prevention of alert fatigue and misuse

---

5. Explainability & Transparency

- Clear justification for recommendations
- Non-technical explanations for administrators
- Auditability and fairness in outputs

Mandatory Innovation Requirement

Teams must propose **at least one advanced innovation**, such as:

- Reinforcement learning–based optimization
- Multi-objective optimization
- Digital twin modeling
- Agent-based simulations
- Incentive or carbon-credit systems
- Self-learning policy engines

PPT SUBMISSION GUIDELINES

Important Note

This is a **PPT (Ideation) Round only**.

- **No working prototype, codebase, or deployed application is required.**

Evaluation will be based entirely on **solution depth, innovation, feasibility, and clarity**.

Mandatory PPT Sections

The presentation **must include**:

1. Problem Understanding
2. Proposed Solution Overview
3. System Architecture (Conceptual)
4. Data Strategy
5. Algorithms & Intelligence (Conceptual)
6. Policy Simulation Approach
7. Innovation Highlight
8. Sustainability Impact
9. Feasibility & Scalability
10. Future Roadmap

PPT SPECIFICATIONS & RULES

PPT Specifications

- **Format:** PPT / PDF
- **Slides: 5-8** (recommended)
- **Language:** English
- **Submission Mode:** UNSTOP PORTAL
- **Deadline: 27 January 2026, 11:59 PM**

Rules & Regulations

- Team submissions must be original
- Plagiarism will result in immediate disqualification
- One submission per team
- Late submissions will not be accepted